

SUPERCRITICAL CO₂ PRESSURE CONTROL SYSTEM

Technical Design Document

Automated Pressurization and Depressurization Rate Control
for High-Pressure Research Vessel

Project:	scCO ₂ Pressure Control System
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1. EXECUTIVE SUMMARY

This document describes the design of an automated pressure control system for a supercritical CO₂ (scCO₂) research vessel in the Taub Research Group at the University of Michigan. The system will provide constant, programmable pressurization and depressurization rates for materials processing experiments, replacing the current manual valve operation.

Key Specifications:

Parameter	Value
Maximum Pressure	28 MPa (4,000 psi)
Operating Temperature	60°C (above CO2 critical point)
Pressurization Rate	350 kPa/min (constant)
Depressurization Rate	2 MPa/min (constant)
Rate Accuracy	±5-10%
Estimated Cost	\$2,500 - \$3,500
Implementation Time	4-6 weeks

Recommended Approach: A custom-built control system using a motorized needle valve for pressurization control and a PWM-controlled solenoid for depressurization. This approach provides 80% of commercial system performance at 20% of the cost, with full data logging capability and complete customization flexibility.

2. SYSTEM REQUIREMENTS

2.1 Functional Requirements

ID	Requirement	Priority
FR-1	Maintain constant pressurization rate of 350 kPa/min $\pm 10\%$	Critical
FR-2	Maintain constant depressurization rate of 2 MPa/min $\pm 10\%$	Critical
FR-3	Hold pressure at setpoint $\pm 1\text{-}2\%$ indefinitely	Critical
FR-4	Operate at vessel temperature of 60°C	Critical
FR-5	Log pressure vs time data to file	High
FR-6	Provide user interface for setpoint entry	High
FR-7	Display real-time pressure reading	High
FR-8	Allow manual override capability	Medium

2.2 Safety Requirements

ID	Requirement
SR-1	Software overpressure limit at 28 MPa (shall not command higher)
SR-2	Mechanical relief valve at 30 MPa (independent of electronics)
SR-3	Hardwired E-stop that cuts power to all actuators
SR-4	Fail-safe on power loss: booster stops, solenoid closes, vessel holds
SR-5	Independent analog pressure gauge for operator verification
SR-6	Check valve to prevent backflow from vessel to booster

3. EXISTING EQUIPMENT ANALYSIS

The following equipment is already installed and operational in the laboratory. The control system design will integrate with these components.

Component	Make/Model	Specifications	Status
CO2 Cylinder	Cryogenic Gases	Bone Dry, 850 psi	Operational
Cylinder Regulator	Victor Lab Master	0-800 psi output	Operational
Air-Driven Booster	Hydraulics Int'l SG-TD-28/150	28 MPa max, 150:1 ratio	Operational
Air Filter/Regulator	Generic (black)	0-150 psi drive air	Operational
HiP Needle Valve	HiP 15-11AF1	10,000 psi, manual	To be motorized
Pressure Vessel	Custom SS	28 MPa rated, multi-port	Operational
Pressure Gauge	Duro United	0-5000 psi analog	Operational
Temperature Controller	INKBIRD	60°C setpoint	Operational
Hot Plate	Thermo Scientific	Vessel heating	Operational
HP Flexible Tubing	Swagelok	1/4" OD, 10,000 psi	Operational

Key Finding: The existing HiP needle valve at the booster output is the ideal location for flow control. By motorizing this valve (or replacing it with a similar motorized unit), we achieve direct control of CO₂ flow into the vessel - the most precise approach for rate control.

4. PROPOSED SYSTEM ARCHITECTURE

4.1 Architecture Overview

The control system uses a closed-loop feedback architecture with direct CO₂ flow control. A pressure transducer provides real-time feedback to a PID controller, which modulates a motorized needle valve (for pressurization) or PWM solenoid (for depressurization) to maintain the desired pressure ramp rate.

4.2 Control Philosophy

Pressurization Control:

The motorized needle valve is positioned between the booster output and the vessel inlet. The controller generates a linear pressure setpoint ramp: $P(t) = P_{\text{start}} + (dP/dt) \times t$. The PID algorithm adjusts valve position to minimize the error between setpoint and measured pressure.

Depressurization Control:

A high-pressure solenoid valve in the vent line operates in PWM mode. By varying the duty cycle (0-100%) at 1 Hz frequency, the controller modulates the average vent rate. A manual needle valve downstream limits maximum flow for safety.

Hold Mode:

During pressure hold, the controller switches to setpoint regulation. The motorized valve makes small adjustments to compensate for any leakage or temperature-induced pressure changes.

5. P&ID SCHEMATIC

5.1 Process Flow Diagram

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■ ■ CONTROLLER ■ ■

4 of 4

■ ■ OUTPUTS: Stepper (PUL/DIR), Solenoid (PWM via SSR) ■ ■

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■ <■■ Signal Flow ■ NEW = Components to Add ■

12345678910111213141516171819202122232425262728293031323334353637383940414243444546474849505152535455565758596061626364656667686970717273747576777879808182838485868788899091929394959697989910010110210310410510610710810911011111211311411511611711811912012112212312412512612712812913013113213313413513613713813914014114214314414514614714814915015115215315415515615715815916016116216316416516616716816917017117217317417517617717817918018118218318418518618718818919019119219319419519619719819920020120220320420520620720820921021121221321421521621721821922022122222322422522622722822923023123223323423523623723823924024124224324424524624724824925025125225325425525625725825926026126226326426526626726826927027127227327427527627727827928028128228328428528628728828929029129229329429529629729829930030130230330430530630730830931031131231331431531631731831932032132232332432532632732832933033133233333433533633733833934034134234334434534634734834935035135235335435535635735835936036136236336436536636736836937037137237337437537637737837938038138238338438538638738838939039139239339439539639739839940040140240340440540640740840941041141241341441541641741841942042142242342442542642742842943043143243343443543643743843944044144244344444544644744844945045145245345445545645745845946046146246346446546646746846947047147247347447547647747847948048148248348448548648748848949049149249349449549649749849950050150250350450550650750850951051151251351451551651751851952052152

5.2 Component Identification

Tag	Description	Type	Location
V-1	CO2 Cylinder Valve	Manual	Gas cylinder
PRV-1	Cylinder Regulator	Victor Lab Master	On cylinder
P-1	Air-Driven Booster	Hydraulics Int'l	Bench
FCV-1	Motorized Needle Valve	HiP + Stepper	Booster output
CV-1	Check Valve	Swagelok	After FCV-1
PV-1	Pressure Vessel	Custom SS	On hot plate
PT-1	Pressure Transducer	Omega PX309	Vessel port
PG-1	Pressure Gauge	Duro United	Vessel manifold
SV-1	Vent Solenoid	Clark Cooper	Vent line
V-2	Vent Needle Valve	Swagelok	After SV-1
PSV-1	Relief Valve	Existing	Vessel port
TE-1	Temperature Element	Thermocouple	In vessel
TIC-1	Temperature Controller	INKBIRD	Control panel

6. BILL OF MATERIALS

6.1 New Components Required

Item	Description	Part Number	Qty	Unit Cost	Total
1	Motorized Needle Valve Assy	HiP + NEMA17 + coupling	1	\$800-1,200	\$800-1,200
2	HP Solenoid Valve (24VDC)	Clark Cooper EH40-04-D024-XP	1	\$600-700	\$600-700
3	Pressure Transducer 0-35MPa	Omega PX309-5KGI	1	\$300-350	\$300-350
4	HP Needle Valve (vent)	Swagelok SS-1RS4-A	1	\$150-180	\$150-180
5	Check Valve	Swagelok SS-4C-1/3	1	\$80-100	\$80-100
6	HP Tee Fittings	Swagelok SS-400-3	2	\$30	\$60
7	HP Tubing 1/4" OD	Swagelok 316SS	20 ft	\$5/ft	\$100
8	Misc HP Fittings	Various Swagelok	lot	-	\$150
9	Raspberry Pi 4 (4GB)	RPi4-4GB	1	\$55	\$55
10	ADS1115 ADC Module	Adafruit	1	\$15	\$15
11	MCP4725 DAC Module	Adafruit	1	\$5	\$5
12	TB6600 Stepper Driver	Generic	1	\$20	\$20
13	NEMA 17 Stepper Motor	2A, 1.8°/step	1	\$15	\$15
14	Flexible Coupling 5mm-6mm	Aluminum	1	\$8	\$8
15	24V Power Supply 60W	Mean Well MDR-60-24	1	\$25	\$25
16	Solid State Relay	Crydom D2425	1	\$30	\$30
17	250 Ohm Resistor (1%)	4-20mA to 1-5V	2	\$2	\$4
18	E-Stop Button (NC)	Panel mount	1	\$15	\$15
19	Project Enclosure	Plastic, DIN rail	1	\$30	\$30
20	Wiring, Connectors	Various	lot	-	\$50

SUBTOTAL: \$2,488-\$2,957

Contingency (15%): \$373-\$443

TOTAL: \$2,861-\$3,400

6.2 Alternative: LabVIEW Controller (+\$300-400)

If using NI DAQ instead of Raspberry Pi, replace items 9-11 with:

- NI USB-6001 DAQ (~\$400) — includes analog inputs, outputs, and digital I/O
- LabVIEW software — check if UMich has campus license (likely yes)

This increases total to approximately \$3,100-\$3,700.

7. CONTROL SYSTEM DESIGN

7.1 Controller Options Comparison

Feature	Raspberry Pi + Python	NI DAQ + LabVIEW
Hardware Cost	~\$100	~\$400-500
Software Cost	Free (Python)	Free (UMich license)
Ease of Setup	Moderate	Easy (built-in)
PID Control	simple-pid library	Built-in PID toolkit
Data Logging	CSV (custom)	TDMS (native)
GUI Development	Tkinter/PyQt	Front panel (easy)
Real-time Performance	Soft RT (~1ms)	Better (DAQmx)
Industry Standard	No	Yes
Learning Curve	Programming required	Visual programming
Handoff to Others	Code documentation	Self-documenting

Recommendation: Start with Raspberry Pi for maximum flexibility and lowest cost. The Python codebase can be easily modified for future experiments. If the department already has NI DAQ hardware available, LabVIEW may be preferable for maintainability.

7.2 PID Control Strategy

The control system uses a proportional-integral-derivative (PID) algorithm to maintain the desired pressure ramp rate. The key insight is that we control *rate*, not absolute pressure, by tracking a time-varying setpoint.

Setpoint Generator (Ramp Function):

```
P_setpoint(t) = P_start + (dP/dt_target) × t
```

Where:

P_start = initial pressure when ramp begins

dP/dt_target = 350 kPa/min (pressurize) or -2 MPa/min (depressurize)

t = time since ramp started

PID Algorithm:

```
error = P_setpoint - P_measured
```

```
integral += error × dt
```

```
derivative = (error - error_prev) / dt
```

```
output = Kp × error + Ki × integral + Kd × derivative
```

Initial Tuning Parameters:

Kp = 0.5 (increase if sluggish, decrease if oscillating)

Ki = 0.1 (disable during ramps, enable for hold mode)

Kd = 0.05 (damping, reduce if response is noisy)

Sample time = 100 ms (10 Hz control loop)

7.3 Software Architecture

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8. WIRING DIAGRAMS

8.1 Raspberry Pi Wiring

```
■ RASPBERRY PI WIRING DIAGRAM ■
■
■ PRESSURE TRANSDUCER (4-20mA, 2-wire) ■
■ Transducer (+) ■
■ 250 Ohm ■
■ Resistor ■
■ ADS1115 A0 ■
■ Transducer (-) ■ GND ■
■
■ Result: 4mA → 1V, 20mA → 5V (use ADS1115 ±4.096V range) ■
■
■ ADS1115 ADC (I2C) ■
■
■ ADS1115 VDD ■ RPi 3.3V ■
■ ADS1115 GND ■ RPi GND ■
■ ADS1115 SDA ■ RPi GPIO 2 (SDA) ■
■ ADS1115 SCL ■ RPi GPIO 3 (SCL) ■
■ ADS1115 ADDR ■ GND (0x48) ■
■
■ TB6600 STEPPER DRIVER ■
■
■ TB6600 PUL+ ■ RPi GPIO 17 ■
■ TB6600 PUL- ■ RPi GND ■
■ TB6600 DIR+ ■ RPi GPIO 27 ■
■ TB6600 DIR- ■ RPi GND ■
■ TB6600 ENA+ ■ RPi GPIO 22 ■
■ TB6600 ENA- ■ RPi GND ■
■ TB6600 A+/A-/B+/B- ■ Stepper Motor ■
■ TB6600 VCC/GND ■ 24V Power Supply ■
■
■ PWM SOLENOID (via SSR) ■
■
■ RPi GPIO 18 ■
■
■ SSR ■
■ RPi GND ■ Input ■
■
■ 24V Supply (+) ■ Output ■
■ Solenoid (+) ■
■ Solenoid (-) ■ 24V Supply (-) ■
■
■ E-STOP (Normally Closed) ■
■
```

```
■ RPi 3.3V ■■■■■■■■■■■■■■■■■■■ E-Stop Button ■■■■■ RPi GPIO 4 ■
```

■ (Configure GPIO 4 with internal pull-down) ■

■ Normal: GPIO 4 = HIGH | Pressed: GPIO 4 = LOW ■

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8.2 NI USB-6001 Wiring

■ NI USB-6001 WIRING DIAGRAM ■

■ USB-6001 PINOUT ■

■ Pin 1: GND Pin 17: P0.0 (D0 - Stepper PUL) ■
■ Pin 2: AI0 (Pressure In) Pin 18: P0.1 (D0 - Stepper DIR) ■
■ Pin 3: AI1 Pin 19: P0.2 (D0 - Stepper ENA) ■
■ Pin 4: GND Pin 20: P0.3 (D0 - Solenoid PWM) ■
■ Pin 5: AI2 Pin 21: P0.4 ■
■ Pin 6: AI3 Pin 22: P0.5 ■
■ Pin 7: GND Pin 23: P0.6 ■
■ Pin 8: AI4 Pin 24: P0.7 ■
■ Pin 9: AI5 Pin 25: P1.0 (DI - E-Stop) ■
■ Pin 10: GND Pin 26: P1.1 ■
■ Pin 11: AI6 Pin 27: P1.2 ■
■ Pin 12: AI7 Pin 28: P1.3 ■
■ Pin 13: GND Pin 29: +5V ■
■ Pin 14: A00 Pin 30: GND ■
■ Pin 15: A01 Pin 31: +2.5V REF ■
■ Pin 16: GND Pin 32: PFI 0 ■

■ CONNECTIONS ■

■ Transducer (+) ■■■ 250Ω ■■■ Pin 2 (AI0) ■
■ Transducer (-) ■■■ Pin 1 (GND) ■
■ Pin 17 (P0.0) ■■■ TB6600 PUL+ ■
■ Pin 18 (P0.1) ■■■ TB6600 DIR+ ■
■ Pin 19 (P0.2) ■■■ TB6600 ENA+ ■
■ Pin 30 (GND) ■■■ TB6600 PUL-/DIR-/ENA- ■
■ Pin 20 (P0.3) ■■■ SSR Input (+) ■
■ Pin 30 (GND) ■■■ SSR Input (-) ■
■ E-Stop NC ■■■ Pin 25 (P1.0) ■
■ E-Stop COM ■■■ Pin 29 (+5V) ■

9. SAFETY ANALYSIS

9.1 Hazard Identification

Hazard	Consequence	Likelihood	Severity	Mitigation
Overpressure	Vessel rupture, projectiles	Low	Critical	Relief valve, SW limits
Rapid depress.	Dry ice, cold burns	Medium	High	Controlled vent rate
Fitting failure	HP gas jet	Low	High	Torque specs, inspection
CO2 accumulation	Asphyxiation	Medium	Critical	Fume hood, CO2 monitor
Electrical shock	Injury	Low	Medium	Grounding, enclosure
Hot surfaces	Burns	Medium	Low	Insulation, warnings
Control failure	Uncontrolled press.	Low	High	E-stop, watchdog

9.2 Safety Interlocks

Hardware Interlocks (Independent of Software):

- Mechanical relief valve on vessel (30 MPa set pressure)
- Hardwired E-stop button that cuts 24V power to all actuators
- Check valve prevents backflow from vessel to booster
- Manual needle valve limits maximum vent rate

Software Interlocks:

- Maximum setpoint limit: 28 MPa (cannot command higher)
- Watchdog timer: if main loop stalls >1 second, stop all actuators
- Rate limiter: reject setpoint changes that would exceed safe ramp rate
- Startup check: verify pressure reading is valid before enabling control

9.3 Fail-Safe Analysis

Failure Mode	System Response	Result
Power loss	Booster stops, solenoid closes	Vessel holds pressure (safe)
Controller crash	Watchdog triggers, outputs to safe state	Vessel holds pressure
Sensor failure	Invalid reading detected, alarm, stop	Manual intervention required
Stepper failure	Valve position frozen	Pressure drift, manual override
Solenoid failure	Remains closed (NC type)	Cannot vent, manual valve backup
Communication loss	Watchdog triggers	System enters safe hold

10. IMPLEMENTATION TIMELINE

Week	Phase	Tasks	Deliverable
1	Procurement	Submit POs, order all components, verify shop air	Components ordered
2	Mechanical	Install valves, fittings, tubing; pressure test with N ₂	Piping complete
3	Electrical	Wire controller, sensors, actuators; verify signals	Wiring complete
4	Software	Develop control code, GUI; bench test with air	Software functional
5	Commission	Low-pressure CO ₂ tests; initial PID tuning	Basic operation
6	Validation	Full-pressure tests; fine-tune PID; document	System operational

Critical Path Items:

- Motorized needle valve assembly — order Week 1 (may have 1-2 week lead time)
- HP solenoid valve — order Week 1 (verify in stock)
- Pressure transducer — order Week 1

Parallel Activities:

- Software development can begin in Week 1 (simulator mode)
- Control enclosure assembly can proceed while waiting for HP components

11. COMMERCIAL VS DIY COMPARISON

Factor	Commercial (Emerson TESCOM)	DIY System
Total Cost	\$12,000 - \$15,000	\$2,500 - \$3,500
Timeline	3-4 months (including lead times)	4-6 weeks
Rate Accuracy	$\pm 2-3\%$	$\pm 5-10\%$
Data Logging	Limited (proprietary format)	Full (custom CSV/TDMS)
Customization	Limited to ERTune parameters	Unlimited (source code)
Technical Support	Emerson applications engineering	Self + online community
Spare Parts	Emerson/TESCOM only	Standard industrial components
Learning Value	Minimal	Significant (controls, programming)
Upgrade Path	Emerson ecosystem	Open platform, any direction
Risk	Low (proven product)	Medium (integration effort)

Recommendation: The DIY system is the preferred approach for this application. The $\pm 5-10\%$ rate accuracy is sufficient for the research requirements (natural fiber composite processing is not sensitive to small rate variations). The 80% cost savings (\$9,500-\$12,000) can be applied to other research needs, and the project provides valuable learning in process control.

When to Consider Commercial:

- If rate accuracy better than $\pm 3\%$ is required
- If regulatory compliance requires certified equipment
- If no personnel are available for integration and maintenance
- If timeline cannot accommodate any development risk

12. SUPPLIER INFORMATION

Component	Supplier	Contact	Notes
HP Solenoid Valve	Clark Cooper (Magnatrol)	(973) 887-6400	clarkcooper.com
E/P Regulator	SMC Corporation	(800) 762-7621	smcusa.com
Pressure Transducer	Omega Engineering	(800) 826-6342	omega.com
HP Fittings	Swagelok (Michigan Fluid Power)	(734) 213-8000	Local pickup available
HP Tubing	Swagelok	Same as above	316SS 1/4" OD
NI DAQ	National Instruments	(888) 280-7645	ni.com, check UMich vendor
Raspberry Pi	Digi-Key / Adafruit	-	digikey.com, adafruit.com
Stepper Motors	StepperOnline	-	omc-stepperonline.com
Stepper Drivers	Amazon / eBay	-	TB6600 generic
Power Supplies	Digi-Key / Mouser	-	Mean Well MDR series
SSR	Digi-Key	-	Crydom D2425
CO2 Gas	Cryogenic Gases	(313) 835-5513	Current supplier

Procurement Notes:

- Check UMich purchasing for vendor agreements (may get discounts on NI, Swagelok)
- Michigan Fluid Power (Swagelok) has local inventory — can pick up same day
- Clark Cooper HP solenoids may have 1-2 week lead time — order first
- Get quotes before submitting POs — prices may vary from estimates